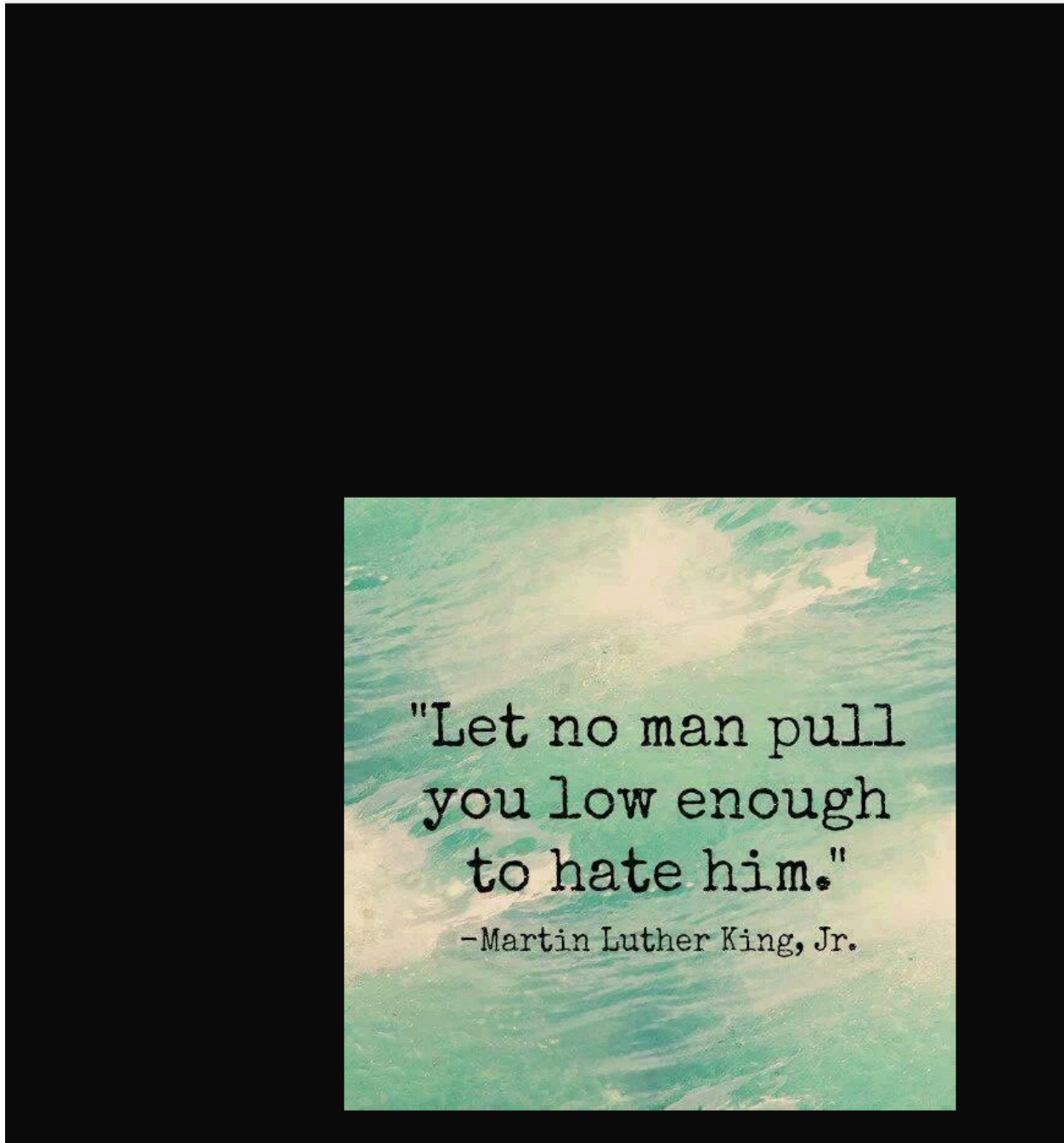
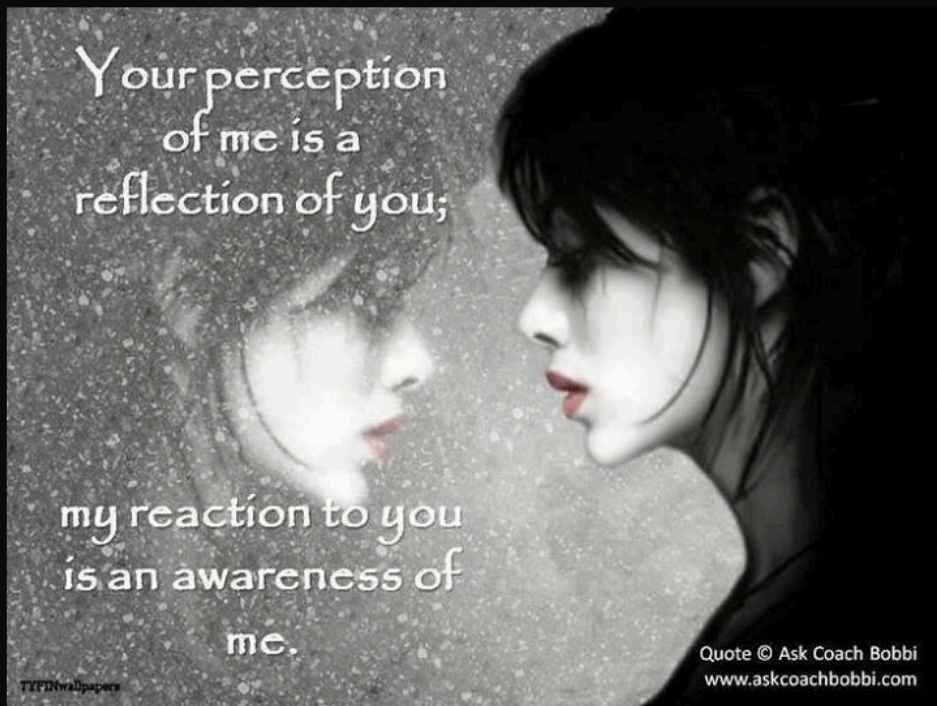


- Take a screenshot of the resulting page including the URL bar.

 <https://chi-ni.com/motd/14.jpg>



- Click "Reload" in the browser and take another screenshot showing the image has changed:



- Use `curl` on your Linux VM to access the API endpoint and show the results. Take a screenshot for your lab notebook.

```
jgafron@guestbook2:~$ curl https://id947kyeff.execute-api.us-east-1.amazonaws.com/default/gettime-jgafron
{"currentTime": "2025-02-26 01:41:36.463904"}jgafron@guestbook2:~$
```

- Take a screenshot that shows that you can view the entries in the backend database.

Guestbook

Name:

Email:

Message:

Sign

Entries

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 04:11:05.517491
Hello DynamoDB!

- Take a screenshot showing that the submission worked.

① /entry - POST method test results		
Request	Latency ms	Status
/entry	670	200
Response body		
[{"message": "Hello DynamoDB!", "date": "2025-02-06 04:11:05.517491", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello Docker DynamoDB!", "date": "2025-02-06 04:39:00.827050", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello Elastic Beanstalk!", "date": "2025-02-12 01:30:27.282449", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello ECS!", "date": "2025-02-12 07:05:30.149379", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello API Gateway", "date": "2025-02-26 07:52:52.783601", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello API Gateway!", "date": "2025-02-26 07:54:11.645992", "email": "jgafron@pdx.edu", "name": "Joseph Gafron"}, {"message": "Hello Cloud9!", "date": "2025-02-06 05:11:37.872590", "email": "joeygafron@gmail.com", "name": "Joseph Gafron"}, {"message": "Hello EC2!", "date": "2025-02-06 05:24:18.392830", "email": "joeygafron@gmail.com", "name": "Joseph Gafron"}]		
Response headers		
{ "Access-Control-Allow-Origin": "***", "X-Amzn-Trace-Id": "Root=1-67bec8a3-19b8b9eca0c841ada9bac22d;Parent=5f2e2776ffe6fc63;Sampled=0;Lineage=1:438eabcf:0" }		

- Take a screenshot as before that shows your entry and the static website hosting URL.

e jgafron-frontend.s3-website-us-east-1.amazonaws.com

Name:

Email:

Message:

Entries

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-06 04:11:05.517491
 Hello DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-06 04:39:00.827050
 Hello Docker DynamoDB!

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-12 01:30:27.282449
 Hello Elastic Beanstalk!

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-12 07:05:30.149379
 Hello ECS!

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-26 07:52:52.783601
 Hello API Gateway

Joseph Gafron <jgafron@pdx.edu>
 signed on 2025-02-26 07:54:11.645992
 Hello API Gateway!

Joseph <jgafron@pdx.edu>
 signed on 2025-02-26 08:26:23.958116
 Hello S3, API Gateway, Lambda!

- Include the hostname for the API gateway in your lab notebook.

gbapigw-6cxjzqx6.uc.gateway.dev

- Take a screenshot of its output

```
...     first_entry = response_json[0]
...     print(f"First entry - Name: {first_entry.get('name','NA')}, Email: {first_entry.get('Email','NA')}, Date: {first_entry.get('date','NA')}, Message: {first_entry.get('message','NA')}")
... except ValueError:
...     print("Error: Response not in JSON")
...
>>> inspect_response()
Response code: 200
Response headers: {'content-type': 'application/json', 'access-control-allow-origin': '*', 'X-Cloud-Trace-Context': '403e613b94de5450e6343697+00:00', 'date': '2025-02-06 07:19:17.908081+00:00', 'message': 'Hello Cloud Shell!', {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-06 04:11:05.517491', 'message': 'Hello DynamoDB!'}, {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-06 04:39:00.827050', 'message': 'Hello Docker DynamoDB!'}, {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-12 01:30:27.282449', 'message': 'Hello Elastic Beanstalk!'}, {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-12 07:05:30.149379', 'message': 'Hello ECS!'}, {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-26 07:52:52.783601', 'message': 'Hello API Gateway'}, {'name': 'Joseph Gafron', 'email': 'jgafron@pdx.edu', 'date': '2025-02-26 07:54:11.645992', 'message': 'Hello API Gateway!'}, {'name': 'Joseph', 'email': 'jgafron@pdx.edu', 'date': '2025-02-26 08:26:23.958116', 'message': 'Hello S3, API Gateway, Lambda!'}]
Data type: <class 'list'>
First entry - Name: Joseph Gafron, Email: NA, Date: 2025-02-06 07:19:17.908081+00:00, Message: Hello Compute Engine!
>>>
```

- Take a screenshot of the output for your lab notebook

```
>>> response = requests.post(url, headers=headers, data=json.dumps(mydict))
>>> print(f"Response code: {response.status_code}")
Response code: 200
>>> print(f"Response Json: {response.json()}")
Response Json: [{'name': 'Joseph Grafon', 'email': 'joeygrafron@gmail.com', 'date': '2025-02-06 07:19:17.908081+00:00', 'message': 'Hello Compute Engine'}, {'name': 'joeygrafron@gmail.com', 'date': '2025-02-06 06:53:06.106613+00:00', 'message': 'Hello Cloud Shell'}, {'name': 'Joseph Grafon', 'email': 'jgafra@43697+00:00', 'message': 'Hello App Engine'}, {'name': 'Joseph Grafon', 'email': 'joeygrafron@gmail.com', 'date': '2025-02-06 07:19:02.526088+00:00', 'name': 'Joey', 'email': 'jgafra@pdx.edu', 'date': '2025-02-27 20:43:46.625224+00:00', 'message': 'Hello Cloud Functions from Python Requests'}, {'name': '@gmail.com', 'date': '2025-02-21 01:10:03.185674+00:00', 'message': 'Hello Kubernetes'}]]
>>>
```

- Take a screenshot showing the preflight request to the API that allows API access, as well as the subsequent fetch request have been successful. Note that this request may not appear in Firefox browsers.

Name	Status	Type	Initiator	Size	Time
 index.html	200	document	Other	844 B	
 style.css	200	stylesheet	index.html:5	566 B	
 guestbook.js	200	script	index.html:6	2.3 kB	
 entries	200	fetch	guestbook.js:67	396 B	1
 entries	200	preflight	Preflight	0 B	1

- **Take a screenshot of the Guestbook including the URL.**

Joseph Gafron <joeegafron@gmail.com>
signed on 2025-02-06 07:19:17.908081+00:00
Hello Compute Engine!

Joseph Gafron <joeegafron@gmail.com>
signed on 2025-02-21 01:39:29.565390+00:00
Hello Cloud Build!

Joseph Gafron <joeegafron@gmail.com>
signed on 2025-02-06 06:53:06.106613+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:41:06.371140+00:00
Hello Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-12 01:48:46.343697+00:00
Hello App Engine!

Joseph Gafron <joeegafron@gmail.com>
signed on 2025-02-06 07:19:02.526088+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:42:58.746814+00:00
Hello Docker Datastore!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-27 22:13:07.420662+00:00
Hello API Gateway from local SPA!

- Take a screenshot of the Guestbook including the URL.

Guestbook

Name:

Email:

Message:

Hello API gateway from SPA in GCS!

Entries

Joseph Gafron <joeygaftron@gmail.com>
signed on 2025-02-06 07:19:17.908081+00:00
Hello Compute Engine!

Joseph Gafron <joeygaftron@gmail.com>
signed on 2025-02-21 01:39:29.565390+00:00
Hello Cloud Build!

Joseph Gafron <joeygaftron@gmail.com>
signed on 2025-02-06 06:53:06.106613+00:00
Hello Cloud Shell!

Joseph Gafron <jgafron@pdx.edu>
signed on 2025-02-06 06:41:06.371140+00:00
Hello Datastore!

Joseph <jgafron@pdx.edu>
signed on 2025-02-27 22:37:13.475609+00:00
Hello API gateway from SPA in GCS!

- Take a screenshot of the Headers that includes the URL and the returned HTTP status code for the first two requests for your lab notebook.

The screenshot shows the Chrome DevTools Headers tab for the first request. The 'Name' column lists resources: style.css, gcp-oauth-gb-498335336970..., style.css, sign, auth?response_type=code&cli..., identifier?opparams=%253F&..., m=_b_tp, and data:image/svg+xml;. The 'Headers' tab is selected, showing the 'General' section. The 'Request URL' is https://gcp-oauth-gb-498335336970.us-west1.run.app/sign. The 'Request Method' is GET. The 'Status Code' is 302 Found. The 'Remote Address' is [2001:4860:4802:34::35]:443. The 'Referrer Policy' is strict-origin-when-cross-origin. The 'Response Headers' section is collapsed.

The screenshot shows the Chrome DevTools Headers tab for the second request. The 'Name' column lists resources: style.css, gcp-oauth-gb-498335336970..., style.css, sign, auth?response_type=code&cli..., identifier?opparams=%253F&..., m=_b_tp, data:image/svg+xml;, 4UaRrENHsxJlGDuGoTOlJfC6l..., m=byfTOblJgVmcLEikZe, m=n73qwfSCuOPbJZT63,vfu..., and m=ltDFwFSD8lzhmumy.587. The 'Headers' tab is selected, showing the 'General' section. The 'Request URL' is https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=498335336970-u8fuc08t9mvl4rrrrfr21v8r3ke13ulv.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fgcp-oauth-gb-498335336970.us-west1.run.app%2Fcallback&scope=https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile&state=NUcIXxRxfJsf85v66gKGfyfynEULPpP8&prompt=login. The 'Request Method' is GET. The 'Status Code' is 302 Found. The 'Remote Address' is [2607:8b0:400::54]:443. The 'Referrer Policy' is strict-origin-when-cross-origin. The 'Response Headers' section is collapsed, showing 'Alt-Svc: h3=":443"; ma=2592000 h3-29=":443"; ma=2592000'.

- Based on the description of the source code, what lines of code in our application are responsible for the second request shown?

This code, in sign.py is responsible redirecting the user to the authorization URL

```
class Sign(MethodView):
    def get(self):
        # If client has an OAuth2 token, use it to get their information and render
        # the signing page with it
        if 'oauth_token' in session:
            google = OAuth2Session(client_id, token=session['oauth_token'])
            userinfo = google.get('https://www.googleapis.com/oauth2/v3/userinfo').json()
            return render_template('sign.html', name=userinfo['name'], email=userinfo['email'], picture=userinfo['picture'])
        else:
            # Redirect to the identity provider and ask the identity provider to return the client
            # back to /callback route with the code
            google = OAuth2Session(client_id,
                                   redirect_uri = redirect_callback,
                                   scope = 'https://www.googleapis.com/auth/userinfo.email ' +
```

- Take a screenshot of the permissions you (as a user) are granting the Guestbook access to.

Sign in to gcp-oauth-gb-498335336970.us-west1.run.app

inafron@ndv.edu

By continuing, Google will share your name, email address, language preference, and profile picture with gcp-oauth-gb-498335336970.us-west1.run.app. See gcp-oauth-gb-498335336970.us-west1.run.app's Privacy Policy and Terms of Service.

You can manage Sign in with Google in your [Google Account](#).

- Take a screenshot of the Headers that includes the entire Callback URL and its returned HTTP status code. What is the URI for the Location that the User sent to by the Callback?

	Headers	Payload	Preview	Response	Initiator	Timing	Cookies
▼ General							
Request URL:	https://gcp-oauth-gb-498335336970.us-west1.run.app/callback?state=K4TFaTnGf3l8TByQ4yp7vxNC5YkXZl&code=4%2F0AQSTgQGOR8tQNTouJgu_I-tf98U1Dya7HxiQhnieUmpC-WxCSF5ueemPJSJOxWyU7kRXjQ&scope=email+profile+openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.profile+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email&authuser=0&hd=pdx.edu&prompt=consent						
Request Method:	GET						
Status Code:	● 302 Found						
Remote Address:	[2001:4860:4802:34::35]:443						
Referrer Policy:	origin						

/sign
Google Frontend

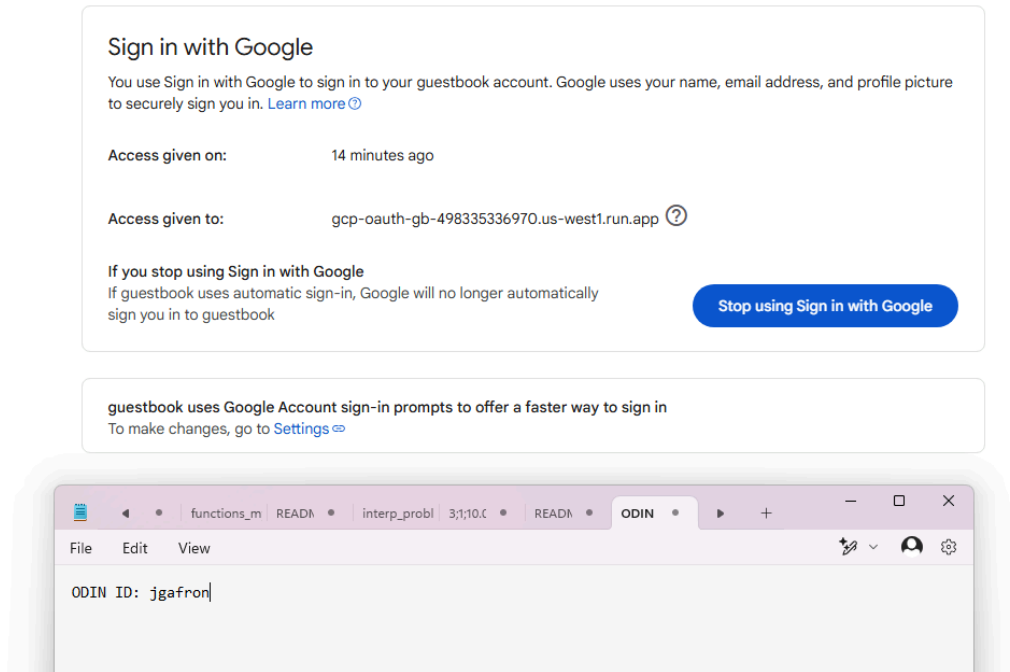
← to the /sign route of the same webpage!

- Find the request within Developer Tools that fetches the embedded image and take a screenshot of its URL.

▼ General	
Request URL:	https://lh3.googleusercontent.com/a/ACg8ocJrZdT8EmYpIFgOQcl3X1cTCfse6Th6NO4JD7Pwo8loRbIVg=s96-c
Request Method:	GET
Status Code:	200 OK
Remote Address:	[2607:f8b0:400a:807::2001]:443
Referrer Policy:	no-referrer
▼ Response Headers	
Access-Control-Allow-Origin:	*
Access-Control-Expose-Headers:	Content-Length
Alt-Svc:	h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
Cache-Control:	public, max-age=86400, no-transform
Content-Disposition:	inline;filename="unnamed.png"
Content-Length:	804
Content-Type:	image/png
Date:	Thu, 27 Feb 2025 23:08:35 GMT
Etag:	"v0"
Expires:	Fri, 28 Feb 2025 23:08:35 GMT
Server:	fife
Timing-Allow-Origin:	*
Vary:	Origin
X-Content-Type-Options:	nosniff
X-Xss-Protection:	0
▼ Request Headers	

- Take a screenshot of the expanded information that includes your OdinId for your lab notebook.

← How Google helps you sign in to guestbook



- What other domains are given access to this Firebase project by default?

Localhost, fir-jgafron.firebaseio.com, fir-jgafron.web.app

- Take a screenshot of the first 10 lines of the produced file.

```
jgafron@cloudshell:~/codelab-friendlychat-web/web-start (firebase-jgafron)$ npm run start

> friendlychat@1.0.0 start
> webpack --watch --progress

asset main.js 4.57 MiB [emitted] (name: main)
runtime modules 617 bytes 3 modules
modules by path ./node_modules/ 1.72 MiB
  modules by path ./node_modules/@firebase/ 1.69 MiB 16 modules
  modules by path ./node_modules/firebase/ 1.17 KiB
    ./node_modules/firebase/app/dist/esm/index.esm.js 827 bytes [built] [code generated]
    ./node_modules/firebase/auth/dist/esm/index.esm.js 70 bytes [built] [code generated]
    ./node_modules/firebase/firestore/dist/esm/index.esm.js 75 bytes [built] [code generated]
    + 3 modules
  modules by path ./node_modules/idb/build/*.js 10.8 KiB
    ./node_modules/idb/build/index.js 3.44 KiB [built] [code generated]
    ./node_modules/idb/build/wrap-idb-value.js 7.32 KiB [built] [code generated]
    ./node_modules/tslib/tslib.es6.mjs 17.2 KiB [built] [code generated]
```

- What missing functions deal with user authentication?

signIn(), signOut(), initFirebaseAuth(), getUsername(), isUserSignedIn()

- What missing functions deal with sending and receiving messages?

saveMessage(), loadMessage(), saveImageMessage(), and checkSignedInWithMessage()

- What are the names of the elements that are hidden when the user is signed out?

userNameElement, userPicElement, SignOutButtonElement

- What is the name of the element that is not hidden when the user is signed out?

signInButtonElement

Include a screenshot of the message and its fields in the database for your lab notebook

Friendly Chat

-  Hello
Joe Gafron
-  HelloHigh
Joe Gafron

 > messages > FeXKkD09I1BIId..

[More in Google Cloud](#)

(default) + Start collection messages >	messages + Add document FeXKkD09I1BIIdmXtA8xn >	FeXKkD09I1BIIdmXtA8xn + Start collection + Add field name: "Joe Gafron" profilePicUrl: "https://lh3.googleusercontent.com/a/ACg8ocJrZdT8EmYpIFgOQc" text: "Hey!" timestamp: March 3, 2025 at 3:17:22 PM UTC-8
---	---	--

Friendly Chat

-  Hey!
Joe Gafron
-  Pretend the instructor added a message
Wu

- **What is the URL of the image that is first shown in the UI as the message is loading?**

LOADING_IMAGE_URL

```
var LOADING_IMAGE_URL = 'https://www.google.com/images/spin-32.gif?a';
```

- **How do the fields in an image document differ from that of the text document?**

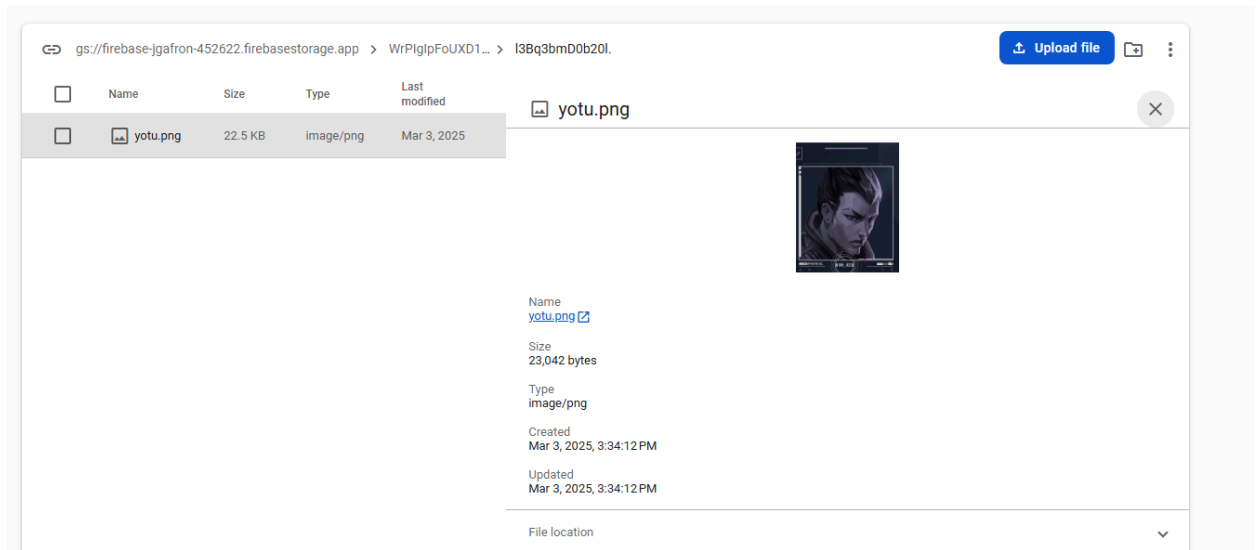
Well there are two additional fields, an Image URL and a storage URI. The image URL is self-explanatory and the the storage URI is the storage location.

- **What URL and storage location can the image be found at?**

"https://firebasestorage.googleapis.com/v0/b/firebase-jgafron-452622.firebaseio.com/o/WrPIgIpFoUXD1AOY0PdUz4ZZof13%2FI3Bq3bmd0b20lrzl6Nlx%2Fyotu.png?alt=media&token=efd222c4-8da7-4010-abb6-9fc01b19ed93"

"WrPIgIpFoUXD1AOY0PdUz4ZZof13/I3Bq3bmd0b20lrzl6Nlx/yotu.png"

- **Take a screenshot of the image in the storage bucket for your lab notebook.**



- What directory is the application going to be served from?

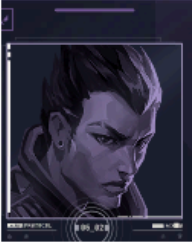
./public

- Take a screenshot of the message including the URL for your lab notebook.

🗨️ Friendly Chat

J Hey!
Joe Gafron

 Pretend the instructor added a message
Wu

J 
Joe Gafron

D Hello Joe!
Dante Sanchez

R my favorite part of the gym is gnawing on this
between sets

Ryan Houlberg

